Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Justification

Code-switching is identified as a HIGH-priority deployment requirement [A2] and is a structural feature of Luxembourgish journalism [WEB-6, WEB-10]. The benchmark explicitly acknowledges that 'most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts' [Q112], and that data are concentrated in a limited set of public domains [Q119]. Only the human-annotated RTL news comment NER corpus is documented as covering 'informal and code-mixed writing' [Q41]; systematic quantification of code-switched content across the other tasks (HA, LA, TC) is absent. RTL is the principal source for HA, SA, TC, and NER comments [Q16, Q24, Q40, Q46], which aligns with the deployment's primary outlet [WEB-1], a meaningful strength. However, LA sentences are sourced from the LOD formal dictionary [Q30], RTE is machine-translated then LLM-improved [Q60, Q61], and intent detection uses German MT [Q54] — meaning much of the benchmark content is not authentic Luxembourgish journalistic text. Given code-switching is a HIGH-priority dimension for this deployment and the benchmark itself flags this as a limitation, score is 2.

ID	Evidence
WEB-6	Luxembourg's three-language administrative regime (Law of 24 February 1984) institutionalises French/German/Luxembourgish co-occurrence in journalistic text
WEB-10	Luxembourgish nationals speak on average 4.3 languages and 9 of 10 residents speak French, confirming code-switching as a structural feature
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity

Strengths

- RTL — the principal Luxembourgish broadcast outlet [WEB-1] — is the dominant source for HA, SA, TC, and NER, providing strong domain alignment for the deployment's primary register [Q16, Q24, Q40, Q46]

ID	Evidence
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity

- The RTL news comment NER subcorpus explicitly covers informal and code-mixed writing [Q41], partially addressing the deployment's code-switching need for NER
- The benchmark explicitly acknowledges its register and code-switching limitations [Q112, Q119], enabling informed deployment risk management

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — Luxembourgish journalism requires handling French, German, and English code-switching, EU institutional terminology, and Grande Région cross-border entities [A2, WEB-1, WEB-10]. The benchmark partially addresses this through the RTL comment NER subcorpus [Q41] but does not systematically cover code-switching across HA, LA, or TC.

ID	Evidence
WEB-10	Luxembourgish nationals speak on average 4.3 languages and 9 of 10 residents speak French, confirming code-switching as a structural feature
WEB-6	Luxembourg's three-language administrative regime (Law of 24 February 1984) institutionalises French/German/Luxembourgish co-occurrence in journalistic text

IC-2: RTL-derived content (HA, SA, TC, NER comments) [Q16, Q24, Q40, Q46] aligns with mainstream Luxembourgish journalistic culture; LA sentences from LOD [Q30] reflect formal/normative register diverging from the deployment's flexible journalistic standard [A3].

IC-3: Intent detection uses xSID voice-assistant commands [Q51, Q55] — a register entirely irrelevant to newsroom deployment. RTE is sourced from English-origin pairs translated and LLM-improved [Q60, Q61], introducing non-Luxembourgish provenance.

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotator audit of code-switched coverage across the benchmark has been performed; gap_id 1 in the regional YAML remains unresolved [NEEDS VERIFICATION]. Direct dataset inspection by Luxembourgish journalism experts would be needed.

IC-5: Documented content issues: (a) systematic under-representation of code-switching outside the NER comment subcorpus [Q112]; (b) LA grounded in formal LOD register [Q30] rather than journalistic register; (c) 22–28% of RTE filtered post-LLM improvement [Q66] indicating quality variance; (d) JUDGEWEL automatically constructed with minimal human verification [Q37].

Information Gaps

- Quantified rate of French/German/English code-switching in HA, LA, and TC instances (gap_id 1, deferred)
- Whether RTL article corpus reflects only published staff articles or includes user-comment register
- Coverage of EU institutional naming conventions and Grande Région entity mentions in the NER training set

Requires Expert Verification

- Whether code-switching density in benchmark instances matches operational density in target newsroom copy
- Whether the RTL-sourced subset reflects the same editorial register as the deploying outlet's output

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Remediation

Gap 1: Code-switching density across HA, LA, and TC tasks is undocumented; only the NER comment subcorpus explicitly covers code-mixed text [Q41, Q112]

Recommendation: Sample 200–500 instances per task and label code-switched tokens; commission a code-switched evaluation supplement drawn from RTL articles to quantify performance under realistic French/German/English mixing